

MINISTRY OF SCIENCE AND HIGHER EDUCATION OF THE RUSSIAN FEDERATION
Federal state autonomous educational institution of higher education “South Ural State
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Institute of Engineering and Technology
Electric drive, mechatronics and electromechanics department
15.04.06 “Mechatronics and Robotics”

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Research on omnidirectional vision system calibration

THESIS WORK OF THE MASTER'S PROGRAM
“MECHATRONICS AND ROBOTICS”
SUSU-15.04.06.2025.265.02.EN TW

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TASK
of thesis work (TW) of the student
of Master's program

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1. The theme of TW of the undergraduate:
Research on omnidirectional vision system calibration
2. The deadline of thesis work: 10.05.2025
3. Contents of the task for explanatory note of thesis:

ANNOTATION

ABSTRACT

INTRODUCTION: With the development of AI, robotics, and IoT, traditional vision has limitations. Omnidirectional vision systems using fisheye lenses are crucial, yet their calibration is challenging due to nonlinear aberrations. This chapter presents the research background, significance, and the status of domestic and international research. Parameter calibration of conventional lens vision systems: Camera calibration is to determine camera parameters and establish coordinate mapping. This chapter introduces the pinhole model, imaging principle, and various calibration methods based on stereo and planar targets, analyzing their advantages and disadvantages.

Fisheye lens imaging model and aberration analysis: Fisheye lenses have ultra-wide angles and short focal lengths. This chapter explains their structure, imaging principle, multiple imaging models, the causes and types of aberrations, and constructs the fisheye vision system model.

Calibration methodology design for OmniVision: This chapter details the fisheye lens calibration method using a checkerboard. It includes the role of the checkerboard, 3D world coordinate generation, corner detection and optimization, calibration function parameters, and the calibration process and parameter interpretation.

Experiments and analysis of results: Experiments are conducted to calibrate the fisheye lens using Python code. The calibration results are obtained, and the distances from the checkerboard to the camera are calculated. The results are compared with those of Matlab to analyze the differences in reprojection error and other aspects.

CONCLUSIONS AND PROSPECTSREFERENCES

4. Slides:

- 1. Research Background and Status
- 2. Parameter Calibration of Conventional Lens Vision Systems
- 3. Imaging Model and Aberration Analysis of Fisheye Lenses
- 4. Calibration methodology design for OmniVision
- 5. Experiments and analysis of results
- 6. Conclusions and Prospects

5. The task is given December 02, 2024

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The task I’ve accepted to execution	_____	T.Zhou

ANNOTATION

T.Zhou. Research on omnidirectional vision system calibration– Chelyabinsk: SUSU, PI, PEF; 2025, 73p., 34 figures, references – 22,29 slides of presentation.

As 360° scene perception grows critical for AI, robotics, and intelligent systems, fisheye-based omnidirectional vision systems—offering single-camera ultra-wide fields ($>180^\circ$)—have become essential, applied in autonomous driving (e.g., Tesla’s surround view), industrial automation, and security. Their strong nonlinear aberrations (radial/tangential distortions), however, challenge traditional calibration: conventional pinhole models and early polynomial fits fail at extreme angles ($>150^\circ$), necessitating specialized frameworks for accurate geometric correction vital to tasks like 3D reconstruction.

Experimental comparison with MATLAB’s toolbox shows its optimized model yields lower reprojection error (0.16 vs. 0.28 pixels), highlighting the impact of model constraints and initial settings. While MATLAB offers superior out-of-the-box precision, the thesis underscores Python-based implementations’ potential through parameter tuning and higher order distortion terms inclusion. Future work focuses on real-time self-calibration, multi-sensor fusion, and lightweight deep learning for edge deployment, aiming to enhance robustness in dynamic environments. This study advances fisheye lens calibration, enabling reliable omnidirectional vision for next-generation intelligent systems.

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ABSTRACT

With the growing demand for 360° scene perception in AI, robotics, and IoT, omnidirectional vision systems based on fisheye lenses have become essential due to their single-camera ultra-wide field of view ($>180^\circ$), finding applications in autonomous driving (e.g., Tesla’s surround view), industrial automation, and security. However, their strong nonlinear aberrations—radial/tangential distortions—pose challenges for traditional calibration methods (e.g., pinhole models, polynomial fitting), which fail at extreme angles ($>150^\circ$). This research analyzes conventional calibration methods (e.g., Tsai’s two-step method, Zhang’s planar target method), constructs a fisheye vision system model based on the equidistant projection principle with detailed aberration analysis, and designs a checkerboard-based calibration methodology involving corner detection, 3D world coordinate generation, and nonlinear optimization in Python. Experiments show that MATLAB’s optimized model yields a lower reprojection error (0.16 pixels vs. Python’s 0.28 pixels), though Python demonstrates potential through higher-order distortion tuning. The study emphasizes the need for real-time self-calibration, multi-sensor fusion, and lightweight deep learning for edge deployment to enhance robustness in dynamic environments.

1. Introduction

1.1 Background and significance

With the rapid development of artificial intelligence, robotics and the Internet of Things, comprehensive perception of the environment and high-precision modeling have become the core needs of intelligent systems on the ground. The traditional vision system is limited by the narrow field of view (usually less than 90°), it is difficult to cover the global information of the complex scene, and Omnidirectional Vision System (Omnidirectional Vision System) realizes 360° dead-angle-free scene perception through the synergy of multiple fisheye cameras or a special optical design, which has become the key technology to solve this problem. At the same time, although the ultra-wide-angle characteristics of fisheye lenses extend the perception range, their complex nonlinear aberration characteristics also bring significant calibration challenges. In the following, we describe the research background from two aspects: technology evolution and application-driven:

Technological evolution: Traditional computer vision relies on monocular or binocular narrow field-of-view cameras, which have a limited perceptual range and require complex calibration and alignment algorithms for multi-camera splicing. The introduction of fisheye lenses enables single-camera field-of-view angles of 180° or more, which in combination with multiple fisheye camera arrays (e.g., quadruple vision systems) or fold-reflective optical designs (e.g., panoramic ring lenses) can achieve full coverage at lower hardware costs. For example, Tesla's Autopilot system builds a body surround view system with four fisheye cameras to generate a bird's eye view in real time to assist with parking and low-speed navigation.

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The traditional pinhole camera model is only applicable to small field-of-view angle scenarios, while the strong nonlinear aberrations (e.g., radial aberrations, centrifugal aberrations) of the fisheye lenses require more complex mathematical model descriptions. Early studies used polynomial fitting of aberrations (e.g., Brown-Conrady model), but the accuracy drops drastically when facing large field of view ($>150^\circ$), prompting researchers to propose fisheye models based on geometric projection principles (e.g., Kannala-Brandt model, Scaramuzza model). However, the robustness of the existing models at extreme field of view angles still needs to be improved.

Traditional calibration relies on a checkerboard calibration board in a laboratory environment, which is difficult to adapt to dynamic scenarios (e.g., parameter drift due to vibration of vehicle-mounted cameras). With the advancement of edge computing and online learning technology, the research focus is gradually shifting to Self-Calibration and real-time parameter updating, which puts forward higher requirements on the efficiency and robustness of calibration algorithms.

Fisheye camera, as an imaging device with ultra-wide field of view (usually up to 180° or even larger), has been widely used in computer vision, robot navigation, intelligent surveillance and other fields in recent years. Its unique imaging characteristics enable it to capture a wide range of scene information that is difficult to be covered by traditional pinhole cameras, but at the same time, the nonlinear aberration introduced by the fisheye lens also brings significant calibration challenges.

There are many application scenarios for fisheye cameras, in terms of automatic driving and intelligent transportation, in the automatic driving system, fisheye cameras are often deployed around the vehicle (such as the surround view system), through a

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single lens to cover the horizontal field of view of nearly 180 °, to monitor the surrounding vehicles, pedestrians and obstacles in real time. For example, the Tesla Autopilot system utilizes a fisheye lens to achieve close-range environmental awareness, effectively making up for the lack of traditional forward-looking cameras in the lateral blind spot. In addition, fisheye cameras can reduce the number of cameras deployed in intelligent traffic monitoring, realize panoramic coverage of intersections through a single camera, and reduce the cost of hardware; in virtual reality and augmented reality, fisheye cameras are able to quickly capture a wide range of scene information and provide panoramic images required for immersive environment reconstruction for VR/AR devices. For example, Microsoft HoloLens utilizes fisheye cameras to capture the user's surroundings in real time, and combines SLAM technology to achieve spatial localization of virtual reality fusion. In film and television production, the fisheye lens is also used to generate 360° panoramic video, providing the audience with an interactive experience of free perspective switching; in industrial automation scenarios, the fisheye camera can be installed at the end of the robotic arm or the drone platform to realize the global monitoring of the complex assembly line or large-scale operation area. For example, a warehouse logistics robot recognizes the location of shelves and plans the path through the fisheye vision system, avoiding the computational burden brought by multi-camera splicing. In addition, in agricultural robots, fisheye cameras can cover a wide area of crop growth monitoring and improve the efficiency of pest and disease identification; in security and panoramic monitoring, fisheye cameras significantly reduce the complexity of multi-camera installation and data synchronization in the traditional monitoring system in the field of security by

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virtue of the advantage of wide-angle coverage of a single camera. For example, bank halls, airport waiting areas and other scenes through the deployment of fisheye cameras to achieve dead-angle monitoring, combined with video analytics algorithms can be real-time detection of abnormal behavior.

Fisheye lens aberration characteristics and the need for calibration: fisheye lens through the non-similarity of the imaging principle (such as equidistant projection, body view projection, etc.) will be three-dimensional space in the large field of view of the light mapped to the two-dimensional image plane, the process leads to image edges of the image appears strong nonlinear geometric aberrations, mainly manifested as:

1) Radial distortion: The degree of distortion increases non-linearly with the distance of the pixel point from the center of the image. Straight lines in the edge area are distorted into curves, and the coefficient of aberration varies significantly with lens design differences.

2) Tangential Aberration: caused by lens manufacturing errors or assembly deviations, resulting in image pixels shifting in the tangential direction, further destroying the realism of the geometric structure.

3) Large field-of-view aberration coupling: when the field-of-view angle exceeds 120° , the linear approximation of the traditional pinhole model fails completely, and the mutual coupling of radial and tangential aberrations makes the complexity of aberration modeling rise sharply.

The need for calibration stems from the following needs:

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1) Geometric Correction: Eliminate aberrations to restore the true geometric structure of the scene to ensure the accuracy of subsequent vision tasks (e.g., target detection, 3D reconstruction).

2) Parameter Acquisition: Accurately calculate the internal parameters (focal length, principal point coordinates) and external parameters (positional attitude) of the camera, which provides the basis for multi-sensor fusion (e.g., joint calibration of LIDAR and camera).

3) Algorithm adaptation: Provide distortion-free input data for deep learning-based vision models to avoid model performance degradation due to distortion interference.

1.2 Current status of domestic and international research

The traditional camera calibration methods are based on pinhole model, and solve the internal parameter (focal length, principal point) and external parameter (positional attitude) of the camera by calibrating the target, which are mainly divided into two categories, one of which is based on the three-dimensional calibrator: the Tsai two-step method (1987): based on the radial alignment constraints (RAC), using the relationship of the co-located points of the calibration plate, solving the linear parameters and then optimizing the coefficients of aberration, which is applicable to low aberration lenses,. However, it is poorly adapted to the large aberrations of fisheye lenses[1]. Zhang Zhengyou calibration method[4] (2000): uses a planar tessellated grid of targets to jointly optimize the internal and external parameters as well as the aberration parameters by means of a single-stress matrix and maximum likelihood estimation. Its flexibility and open source implementations (e.g., OpenCV) make it a mainstream method in industry, but the model needs to be extended to support highly nonlinear aberrations in

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fish-eye lenses. The second is the self-calibration method: calibration is performed by natural features in the scene (e.g., vanishing points, motion information) without the need for a dedicated target (e.g., Maybank-Faugeras algorithm). However, the robustness to sparse features or dynamic scenes is poor, and it is difficult to be directly used in fish-eye systems. Traditional calibration methods mainly rely on physical calibrators (e.g., checkerboard grids, circular arrays) or geometric constraints of the scene, and describe the aberration characteristics and optimize the parameters through mathematical models. Scholars at home and abroad have proposed a variety of classical methods in this field.

I .Current status of research on traditional calibration methods

①Progress in foreign research:

Calibrator-based methods:

Foreign scholars started earlier in fish-eye calibration modeling. The polynomial model (OCamCalib)[5] proposed by Scaramuzza et al. (2006) supports the calibration of field-of-view angles exceeding 180° by fitting fish-eye aberrations through polynomials, and has become a mainstream method in open-source tools. The generalized fish-eye model proposed by Kannala and Brandt[6] (2006) (Kannala-Brandt model) is based on symmetric projection properties and separates the aberration parameters from the internal parameters, which significantly improves the calibration accuracy.

Methods based on geometric constraints:

Hughes et al. (2010) proposed to use straight line features in the scene for calibration, and fit curves in the distorted image to straight lines through inverse optimization to solve the distortion parameters. Urban et al. (2015) further introduced vanishing point

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constraints to optimize the camera parameters in conjunction with multiple sets of straight lines in orthogonal directions, which is suitable for structured scenes (e.g., roads, indoor corridors).

②Progress in domestic research:

Model Improvement and Optimization

Domestic scholars focus on the localized improvement of the model on the traditional calibration method. The Tsinghua University team (2018) proposed a fisheye calibration method based on sub-region fitting, which divides the image into central and edge regions, and adopts different aberration models for parameter optimization, which effectively mitigates the edge aberration fitting error [7].

Low-cost calibration program:

Harbin Institute of Technology (2020) has developed a monocular vision-based online calibration algorithm for fisheye cameras, which requires only a small number of calibration plate images for parameter estimation and is suitable for rapid deployment in industrial sites.

Application Scenario Expansion:

The Institute of Automation, Chinese Academy of Sciences (2021) proposes a joint calibration framework for multiple fisheye cameras for in-vehicle surround view systems, which achieves high-precision calibration of external parameters through feature point matching and global optimization, and has been applied to the autonomous driving platforms of several automobile companies.

The traditional method relies on manually designed calibrators or scene a priori, which is insufficiently robust under complex dynamic environments (e.g., light change,

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occlusion), and the calibration process is time-consuming, which makes it difficult to satisfy real-time requirements.

II .Research status of deep learning based calibration methods

In recent years, deep learning techniques have provided new ideas for fisheye calibration, with the core advantage of end-to-end parameter prediction and environment adaptive capability.

①Progress in foreign research:

End-to-end parametric regression:

The DeepCalib network proposed by Rong et al. (2020) firstly applies convolutional neural networks (CNNs) to fisheye calibration by predicting the focal length and aberration coefficients directly from a single image through multi-scale feature fusion without the need for a calibration board[8].CalibNet (Liao et al., 2021) further introduces spatial transformation networks (STNs) to realize the image-to implicit mapping of ideal pinhole images.

Self-supervised learning:

The Google team (2022) proposed a self-supervised calibration framework based on video sequences, which utilizes the optical flow consistency between neighboring frames to constrain the camera parameters and significantly reduces the dependence on annotation data.

Multimodal fusion:

MIT researchers (2023) combined fisheye imagery with LiDAR point clouds to achieve co-calibration through cross-modal feature alignment, validating its effectiveness in an autonomous driving scenario.

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②Progress in domestic research:

Lightweight network design:

Zhejiang University (2021) proposed a lightweight calibration network, FishCalib, which uses channel pruning and quantization techniques to reduce the model computation to 30% of the traditional method for embedded device deployment[9].

Adversarial Generative Data Enhancement:

The Peking University team (2022) used GAN to generate synthetic fisheye images, combined with domain adaptation techniques to enhance the generalization ability of the model in real scenes, and solved the problem of insufficient training data.

Dynamic environment adaptation:

Shanghai Jiao Tong University (2023) proposes a time-aware calibration network (TAC-Net), which captures the temporal drift characteristics of camera parameters through LSTM modules, and realizes sub-pixel-level calibration accuracy in in-vehicle scenes.

③Challenges of Deep Learning Methods

Insufficient physical interpretability: the parameters predicted by the network lack clear physical meaning, making it difficult to be compatible with traditional models.

Real-time bottleneck: complex network structure leads to computational delays, making it difficult to meet the demands of high frame rate applications.

III.Research status of self-calibration and online calibration technology

Aiming at the problem of camera parameter drift, scholars at home and abroad have carried out in-depth research in the field of self-calibration and online calibration.

①Progress in foreign research:

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Vision SLAM Integration:

ORB-SLAM3 (2020) supports dynamic calibration of fisheye lenses by jointly optimizing camera position and internal reference through bundle adjustment (BA).

Vision-Inertial Fusion:

The ETH Zurich team (2021) proposes the VIO (Visual-Inertial Odometry) framework to update the fisheye camera parameters in real-time by Extended Kalman Filtering (EKF), which achieves good results in UAV navigation.

②Progress in domestic research:

Multi-sensor co-calibration:

The Baidu Apollo team (2022) developed a co-calibration system for multiple fisheye cameras and millimeter-wave radar, using semantic segmentation and point cloud alignment for cross-modal parameter optimization[10].

Edge computing optimization:

Huawei Hesse (2023) proposed an NPU-based online calibration acceleration scheme to compress the calibration elapsed time from milliseconds to microseconds, which has been applied to smart cockpit systems.

IV. Challenges and future trends

①Technical Challenges

Extreme aberration modeling: the nonlinear aberration in the fisheye edge region still lacks a common mathematical model.

Multi-sensor calibration: the joint calibration accuracy of fisheye camera with LiDAR and infrared camera needs to be further improved.

②Future Direction

Interpretable AI and physical model fusion: combining neural networks and traditional optimization methods to improve the credibility of calibration results.

Calibration of event cameras: develop adaptive algorithms for the asynchronous characteristics of new event cameras.

Localized calibration tool chain: China needs to break through the underlying technology dependence of calibration software (e.g. MATLAB toolbox).

Fisheye camera calibration technology has developed rapidly from the traditional geometric method to the intelligent and real-time direction. Foreign countries occupy a leading position in basic theoretical models (such as the Kannala-Brandt model) and open source tools (such as OCamCalib), while the domestic application scenario optimization (such as in-vehicle surround-view system) and lightweight algorithm design excel. In the future, with the growing demand for autonomous driving and meta-universe, high-precision and low-cost calibration technology will become the focus of competition at home and abroad, while multi-modal fusion and edge computing optimization will be the breakthrough direction.

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2. Parameter calibration of conventional lens vision systems

Camera calibration is the process of determining the internal and external parameters of a camera. The purpose of camera calibration is to establish an accurate mapping between image pixel coordinates and 3D spatial coordinates for subsequent computer vision tasks such as accurate 3D reconstruction, attitude estimation, and measurement.

2.1 Camera pinhole model and imaging principle

The camera pinhole model is a simplified model used to describe the imaging principle of a camera, which approximates the imaging process of a camera as a pinhole imaging process.

The model views the camera as a dark box with an ideal pinhole, and light is projected from objects in the scene through the pinhole onto the imaging plane to form an inverted real image. In the pinhole model, the relationship between the optical center of the camera (i.e., the position of the pinhole), the imaging plane, and the object can be described by a mathematical formula.

Usually, we establish a three-dimensional camera coordinate system, with the optical center of the camera as the origin, the direction of the optical axis as the positive direction of the z-axis, the imaging plane is perpendicular to the z-axis, and the x-axis and y-axis are parallel to the horizontal and vertical directions of the imaging plane, respectively. Under this coordinate system, the image point $p(x, y)$ of a point $P(X, Y, Z)$ in space on the imaging plane can be obtained by the similarity triangle principle between them. In order to quantitatively describe the camera imaging process, four reference coordinate systems are introduced: the image coordinate system, the imaging

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plane coordinate system, the camera coordinate system, and the world coordinate system.

I .Visual system coordinate system

① Image coordinate system

The image acquired by the camera is stored in the computer in the form of an array, and each element (pixel) value in the array becomes the brightness of the image point [12]. The image coordinate system is the coordinate system O_0uv established on the digital image, as shown in Figure 2-1 with the vertex of the upper left corner of the image as the origin O_0 , the u -axis is positive to the right, and the v -axis is positive to the downward, and the number of columns and rows of each pixel in the array is expressed in the coordinates (u,v) , so that the image coordinate system coordinates (u,v) are expressed in units of pixels.

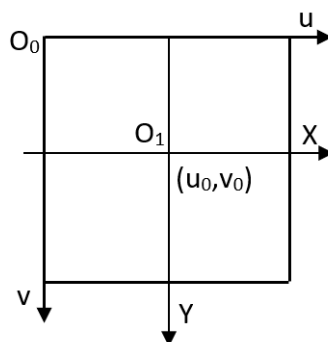


Figure 2-1 Image coordinate system and imaging plane coordinate system

② Image plane coordinate system

Since (u,v) only indicates the number of columns and rows of the pixel located in the digital image, but not expressed in physical units of the pixel's position in the digital image, it is necessary to re-establish the physical units (such as centimeters) expressed in the imaging plane coordinate system O_1XY , the origin of this coordinate system, O_1 ,

is a point in the image, and the two axes, X-axis, Y-axis, respectively, and u, v-axis parallel. Where, (X,Y) is the imaging plane coordinate system coordinates in terms of a certain physical unit length. If O_1 has the coordinates (u_0, v_0) in the O_0uv coordinate system, Dx, dY are the physical dimensions of a single pixel in the direction of the X, Y axes, respectively, then the relationship between the two coordinate systems is:

$$\begin{cases} u = \frac{X}{dX} + u_0 \\ v = \frac{Y}{dY} + v_0 \end{cases} \quad (2-1)$$

For ease of description, the transformation into chi-square coordinates with matrices of the form:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{dX} & 0 & u_0 \\ 0 & \frac{1}{dY} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \quad (2-2)$$

③ Camera coordinate system

Camera imaging geometry is shown in Figure 2-2. Camera on the fixed right-angle coordinate system - camera coordinate system $Oxyz$, the origin O for the optical center of the camera, x, y axis were parallel to the imaging plane coordinate system X, Y axis, z axis is perpendicular to the imaging plane of the camera optical axis. The origin of the imaging plane coordinate system is the intersection of the optical axis and the imaging plane. OO_1 is the focal length f of the camera.

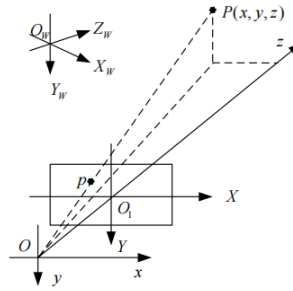


Figure 2-2 Camera Coordinate System and World Coordinate System

④ World coordinate system

The world coordinate system $O_w X_w Y_w Z_w$ is the base coordinate system used to describe the position of any camera and object in the environment, as shown in Figure 2-3. As a reference, this coordinate system is an arbitrary artifactual coordinate system that can be tailored to specific situations. Therefore, in computer vision, the world coordinate system is usually defined in terms of the camera coordinate system or the center 3D coordinate system of the target.

We usually use the translation vector t and the rotation matrix R to represent the relationship between the camera coordinate system and the world coordinate system. Assuming that a point P within the scene has the coordinates $P_w = (X_w, Y_w, Z_w, 1)^T$ and $p = (x, y, z, 1)^T$ in the two coordinate systems, respectively, the relationship is satisfied:

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (2-3)$$

where R is a 3×3 orthogonal unit matrix; t is a three-dimensional translation matrix; and $0 = (0, 0, 0)^T$.

II Camera Pinhole Imaging Model

In the field of computer vision, machine modeling aims to realize the transformation from a 3D scene to a 2D image, and this correspondence can be presented by the following mapping from 3D space to 2D space.

$$f: \begin{pmatrix} R^3 \rightarrow R^2 \\ (x, y, z) \rightarrow (x', y') \end{pmatrix} \quad (2-4)$$

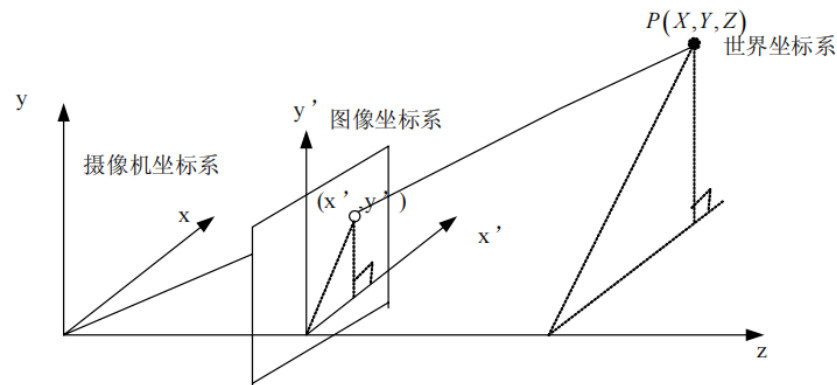


Figure 2-3 Imaging model of a conventional camera

The conventional camera imaging model is shown in Figure 2-3. Any point P in the scene can be projected on the plane image by the camera, and the position of this imaging point in the image can be approximated by the pinhole imaging model. The pinhole imaging model is characterized by the fact that all rays from the scene pass through the optical center. As shown in Figure 2-2, the projection P of a point P in space in the image is the intersection in the image plane of the line OP connecting the center of light O and the point P. The projection P is the intersection of the center of light O and the line OP connecting the center of light O and the point P in the image plane. This relationship is also known as perspective projection or center projection. The following scaling relationships exist:

$$\begin{cases} X = \frac{fx}{z} \\ Y = \frac{fy}{z} \end{cases} \quad (2-5)$$

where (X,Y) denotes the image coordinates of point P, (x,y,z) are the coordinates of point P in the camera coordinate system, and f is the camera focal length, which denotes the distance between the image plane and the xy plane. Translating the above perspective relation into its sub-coordinate form is:

$$s \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad (2-6)$$

where s is a scale factor and H is the perspective projection matrix. Substituting Eqs. (2-2) and (2-3) into the above equation yields the relationship between a point P and its projection p(u,v) in the world coordinate system:

$$\begin{aligned} s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} &= \begin{bmatrix} \frac{1}{dX} & 0 & u_0 \\ 0 & \frac{1}{dY} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \alpha_x & 0 & u_0 & 0 \\ 0 & \alpha_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = M_1 M_2 X_w = M X_w \end{aligned} \quad (2-7)$$

where α_x, α_y denote the scale factor on the u-axis and v-axis, respectively; M is a 3x3 projection matrix; M_1 is determined by the camera internal parameters $\alpha_x, \alpha_y, u_0, v_0$; and M_2 is determined by the camera's orientation with respect to the world coordinate system, which belongs to the camera external parameters.

The pinhole imaging model of a conventional camera is also called a linear model, which is due to the fact that the optical imaging system of the camera described by the

model satisfies the linear transformation relationship. However, in practical application scenarios, the imaging process of the camera is not strictly follow the ideal center projection principle, all types of optical imaging systems will exist in different degrees of optical aberrations. Especially in the optical imaging system composed of wide-angle lenses, this aberration phenomenon is often more significant[18]. Due to these optical aberration errors, the actual image plane coordinates (X',Y') are shifted, and the relationship between the ideal coordinates (X,Y) and the real coordinates (X',Y') is as follows:

$$\begin{cases} X = X' + \delta_x \\ Y = Y' + \delta_y \end{cases} \quad (2-8)$$

Where δ_x , δ_y represent the nonlinear distortion values in the X-axis and Y-axis directions, respectively. Tangential distortion and radial distortion mainly exist in the nonlinear model, and the nonlinear distortion is shown in Figure 2-4.

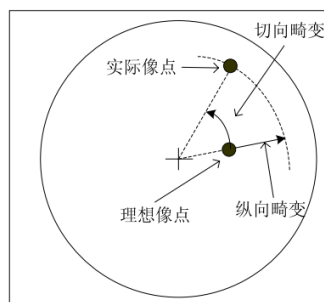


Figure 2-4 Nonlinear distortion

2.2 Based on stereo target calibration method

The calibration technique based on the three-dimensional target is based on the method in photogrammetry, and the theory is relatively mature at present. The calibration technique is to place a three-dimensional target in front of the camera, as shown in Figure 2-5. The vertices of each small square on the target can be used as

feature points, and the internal and external parameters of the camera are solved by the world coordinates and image coordinates of these feature points.

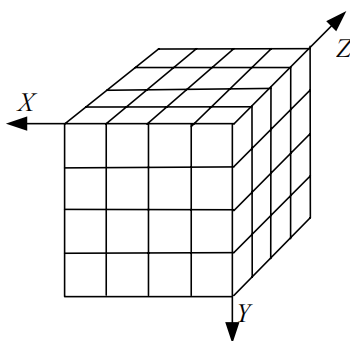


Figure 2-5 3D Stereo Target

This section describes the calibration methods based on perspective transformation matrices as well as linear linear transformation (DLT) calibration methods.

2.2.1 Calibration method based on perspective transformation matrices

In practical application scenarios, the equations used to describe the imaging model of a camera are mostly nonlinear because of optical aberrations in the lens. However, if the nonlinear aberrations are excluded from the analysis of the imaging process, it is possible to solve for the unknown elements of the perspective transformation matrix with the help of a linear method based on a set of known world coordinates of the feature points, as well as the coordinates of the image points corresponding to these feature points[19].

The linear model of the camera is given by:

$$s_i \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X_{wi} \\ Y_{wi} \\ Z_{wi} \\ 1 \end{bmatrix} \quad (2-9)$$

where $(X_{wi}, Y_{wi}, Z_{wi}, 1)$ are the world coordinates of the i th feature point on the target; $(u, v, 1)$ are the coordinates of the image corresponding to it; and m_{ij} are the elements of

the i th row and j th column of the perspective transformation matrix M . Eq. (2-9) contains three equations:

$$\begin{aligned} s_i u_i &= m_{11}X_{Wi} + m_{12}Y_{Wi} + m_{13}Z_{Wi} + m_{14} \\ s_i v_i &= m_{21}X_{Wi} + m_{22}Y_{Wi} + m_{23}Z_{Wi} + m_{24} \\ s_i &= m_{31}X_{Wi} + m_{32}Y_{Wi} + m_{33}Z_{Wi} + m_{34} \end{aligned} \quad (2-10)$$

Eq. (2-10) is collapsed by eliminating s_i to obtain two linear equations for m_{ij} as shown in the following equation:

$$\begin{cases} m_{11}X_{Wi} + m_{12}Y_{Wi} + m_{13}Z_{Wi} + m_{14} - u_i X_{Wi} m_{31} - u_i Y_{Wi} m_{32} - u_i Z_{Wi} m_{33} = u_i m_{34} \\ m_{21}X_{Wi} + m_{22}Y_{Wi} + m_{23}Z_{Wi} + m_{24} - v_i X_{Wi} m_{31} - v_i Y_{Wi} m_{32} - v_i Z_{Wi} m_{33} = v_i m_{34} \end{cases} \quad (2-11)$$

Equation (2-11) indicates that if there are n feature points on the target and the world coordinates $(X_{wi}, Y_{wi}, Z_{wi}, 1)$ ($i=1, 2, \dots, n$) of these points are known as well as the coordinates of the image points corresponding to them (u_i, v_i) ($i=1, 2, \dots, n$), $2n$ linear equations can be obtained as described above, and their matrix form is as follows:

$$\begin{bmatrix} X_{W1} & Y_{W1} & Z_{W1} & 1 & 0 & 0 & 0 & 0 & -u_1 X_{W1} & -u_1 Y_{W1} & -u_1 Z_{W1} \\ 0 & 0 & 0 & 0 & X_{W1} & Y_{W1} & Z_{W1} & 1 & -v_1 X_{W1} & -v_1 Y_{W1} & -v_1 Z_{W1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{Wn} & Y_{Wn} & Z_{Wn} & 1 & 0 & 0 & 0 & 0 & -u_n X_{Wn} & -u_n Y_{Wn} & -u_n Z_{Wn} \\ 0 & 0 & 0 & 0 & X_{Wn} & Y_{Wn} & Z_{Wn} & 1 & -v_n X_{Wn} & -v_n Y_{Wn} & -v_n Z_{Wn} \end{bmatrix} \begin{bmatrix} m_{11} \\ m_{12} \\ m_{13} \\ m_{14} \\ m_{21} \\ m_{22} \\ m_{23} \\ m_{31} \\ m_{32} \end{bmatrix} = \begin{bmatrix} u_1 m_{34} \\ v_1 m_{34} \\ \dots \\ \dots \\ \dots \\ \dots \\ \dots \\ n_{34} \end{bmatrix} \quad (2-12)$$

Since the multiplication of the projection matrix M with any non-zero constant does not affect the relationship between the world coordinates and the image coordinates, the above equation can be taken to be $m_{34} = 1$. Therefore, the other 11 $2n$ linear equations can be obtained with respect to the elements of the matrix M . Noting the 11 unknown elements as the 11-dimensional vectors m , the above equation is simplified to:

$$Km = U \quad (2-13)$$

where K and U are the $2n \times 11$ matrix on the left side and the $2n$ -dimensional vector on the right side of Eq. (2-12), respectively, and are known vectors. When the number of feature points $n \geq 6$, the above linear equation can be solved by the least squares method as:

$$m = (K^T K)^{-1} K^T U \quad (2-14)$$

Since in reality, there are dozens of feature points on the target for calibration, the number of unknown linear equations can be obtained far more than the number of unknowns, so here we use the least-squares method that can reduce the error to solve. The perspective transformation matrix is decomposed to obtain the parameters inside and outside the camera.

2.2.2 Direct Linear Transformation (DLT)

In the early 1870s, Abdel.Aziz and Karara proposed the linear transformation calibration method from the perspective of photogrammetry [20], which constructs a geometric linear model of camera imaging describing the relationship between scene objects and imaging, and completes the solution of the model parameters through linear equations. However, since this method does not consider the nonlinear distortion factors in the camera imaging process, it still needs to improve the calibration accuracy of the camera with the help of nonlinear optimization algorithms. Under the chi-square coordinate system, the DLT method shows the imaging geometric relationship between object points and image points in the form of perspective projection matrix.

The direct linear transformation is modeled as:

$$\begin{cases} u = \frac{X_W l_{00} + Y_W l_{01} + Z_W l_{02} + l_{03}}{X_W l_{20} + Y_W l_{21} + Z_W l_{22} + l_{23}} \\ v = \frac{X_W l_{10} + Y_W l_{11} + Z_W l_{12} + l_{13}}{X_W l_{20} + Y_W l_{21} + Z_W l_{22} + l_{23}} \end{cases} \quad (2-15)$$

Where (X_w, Y_w, Z_w) denotes the three-dimensional coordinates of the feature point on the three-dimensional target; (u, v) denotes the coordinates of the projection point of the feature point on the image coordinate system; l_{ij} is a parameter to be determined (without loss of generality, l_{23} takes the value of 1). If the 3D coordinates (X_w, Y_w, Z_w) of n ($n \geq 6$) feature points on the 3D target are known, as well as the coordinates of their projection points (u, v) in the image coordinate system, the 11 parameters can be calculated by the least squares method. The DLT method is the same as the camera calibration method based on the perspective transformation matrix already discussed above when nonlinear aberrations in the optical imaging process are not considered.

The above is the linear solution part of the linear transformation method [21], the advantage of which is that there is no need for iteration, and the operation speed is faster; the disadvantage is that it does not include the camera system error, and the constraints exist in the intermediate parameters to make the dimension larger than the original parameter dimensions, coupled with the noise, the lens aberration and other influences, the system calibration accuracy is low. When considering the nonlinear distortion, the correspondence between the image points and the feature points on the stereoscopic target in the linear transformation method satisfies the following equation:

$$\begin{cases} u_i + \delta_x(u_i, v_i) = u'_i = \frac{X_{Wi} l_{00} + Y_{Wi} l_{01} + Z_{Wi} l_{02} + l_{03}}{X_{Wi} l_{20} + Y_{Wi} l_{21} + Z_{Wi} l_{22} + l_{23}} \\ v_i + \delta_y(u_i, v_i) = v'_i = \frac{X_{Wi} l_{10} + Y_{Wi} l_{11} + Z_{Wi} l_{12} + l_{13}}{X_{Wi} l_{20} + Y_{Wi} l_{21} + Z_{Wi} l_{22} + l_{23}} \end{cases} \quad (2-16)$$

Where (X_{wi}, Y_{wi}, Z_{wi}) denotes the 3D coordinates of the i th feature point on the target; (u_i, v_i) are the image coordinates of the actual projection points of the feature points (X_{wi}, Y_{wi}, Z_{wi}) in the image coordinate system that can be obtained by digital image processing techniques; (u'_i, v'_i) are the ideal image point coordinates after correction; $\delta_x(u_i, v_i)$ and $\delta_y(u_i, v_i)$ denote the imaging error at the image point (u_i, v_i) , thus it is more convenient to include the nonlinear aberration factor in the DLT method.

According to the above theoretical analysis, the camera calibration system can be simplified to a linear mathematical model when lens distortion is ignored. However, this simplified model is only applicable to small viewing angle imaging scenarios, when there are significant distortion effects of the lens, the image edge region will produce obvious geometric deformation, at this time the linear model can not accurately characterize the actual imaging process. In order to improve the calibration accuracy, it is necessary to establish a nonlinear mathematical model containing aberration parameters, and solve the complete camera parameters through an iterative optimization algorithm. By introducing multi-dimensional correction factors such as radial aberration and tangential aberration, this modeling method can more completely describe the causes of physical aberrations in the optical system, thus significantly improving the calibration accuracy. However, the cost is that the dimensional expansion of the parameter space leads to an exponential increase in the computational complexity and requires a more time-consuming nonlinear optimization algorithm for the solution, which results in a relative reduction in the real-time processing performance of the system.

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2.3 RAC method of R. Tsai

Based on the advantages and shortcomings of the DLT method and the perspective transform matrix method, Tsai proposed a two-stage (Two-Stage) calibration method based on radial constraints in 1986 [2][3]. R. Tsai's RAC method (Radial Alignment Constraint) is one of the classical camera calibration methods, proposed by Roger Y. Tsai in 1987. The method efficiently solves the internal parameters (e.g., focal length, principal points) and radial aberration parameters of the camera by exploiting the radial alignment properties of the feature points in the image and combining them with nonlinear optimization techniques. The core of Roger Tsai's two-step method is based on the fact that the following radial alignment constraint is implemented, which is sometimes shortened to the RAC two-step method. The advantage of the RAC-based method is that the parameter solving equations are mostly linear, which reduces the complexity and makes the method moderately computationally intensive and highly accurate, thus making the calibration process fast and accurate. The following is a detailed description of its core principles, steps and applications:

① Core Ideas

The RAC method is based on the following key assumptions:

Radial aberrations dominate: it is assumed that image aberrations are mainly caused by radial aberrations (e.g., fisheye or barrel aberrations) and tangential aberrations can be ignored.

Dominant point is known or approximately fixed: The dominant point (the intersection of the optical axis and the imaging plane) is assumed to be located near the center of the image, or is predetermined by other methods.

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The calibration plate plane is aligned with the camera coordinate system: the geometric constraints of a planar calibration plate (e.g., a checkerboard grid) are utilized.

By extracting the correspondence between the image coordinates of the corner points of the calibration plate and the world coordinates, combined with the radial alignment constraints, the camera parameters are solved in a step-by-step optimized manner.

②Methodology Steps

Step 1: Data Acquisition

Use a flat calibration board (e.g., a checkerboard grid) to take multiple images at different positions.

Extract the image coordinates (u,v) and the corresponding world coordinates (X_w, Y_w, Z_w) of the corner points of the calibration board (usually set the plane of the calibration board as $Z_w=0$).

Step 2: Solve for external parameters (rotation and translation matrices)

Preliminarily estimate the camera external parameters (rotation matrix R and translation vector T) using Direct Linear Transformation (DLT) or orthogonal projection model.

Step 3: Apply radial alignment constraints

Key constraint: in the absence of aberrations, the projection direction of a calibration plate corner point (the ray from the principal point to the image coordinates of the corner point) should be aligned with the 3D direction under the world coordinate system of that corner point.

Combine the radial aberration model (e.g., k_1, k_2 polynomials) to establish the nonlinear equations and optimally solve for the focal length f and the aberration coefficients.

Step 4: Nonlinear optimization

Jointly optimize all parameters (internal, external, and aberration coefficients) using a least squares method (e.g., Levenberg-Marquardt algorithm) to minimize the reprojection error.

③ Mathematical Model

Radial distortion model

Aberration correction formula for image coordinates:

$$\begin{cases} u_{\text{corrected}} = u_d + (u_d - u_0)(k_1 r^2 + k_2 r^4) \\ v_{\text{corrected}} = v_d + (v_d - v_0)(k_1 r^2 + k_2 r^4) \end{cases} \quad (2-17)$$

where:

(u_d, v_d) are the coordinates of the distorted image, and

(u_0, v_0) are the principal points.

$r = \sqrt{(u_d - u_0)^2 + (v_d - v_0)^2}$ is the radial distance

k_1, k_2 are the radial distortion coefficients.

Projection equation

The projection relation from world coordinates to image coordinates:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \frac{f}{Z_c} \begin{bmatrix} X_c \\ Y_c \end{bmatrix} + \begin{bmatrix} u_0 \\ v_0 \end{bmatrix}$$

Where (X_c, Y_c, Z_c) are the 3D point coordinates in the camera coordinate system.

④ Advantages

Efficient: reduce computational complexity by solving in steps (outer parameter first, then inner parameter).

Robust to noise: suitable for low resolution or noisy images.

Accurate radial distortion correction: specialized for strong radial distortion scenarios such as fisheye lens.

⑤ Limitations

Main point assumption: the main point needs to be pre-fixed or approximated as the center of the image, which may affect the calibration results of high-precision scenes.

Ignore tangential aberration: Not applicable to lenses with significant tangential aberration (e.g. wide-angle lenses).

Depend on the initial value: nonlinear optimization is sensitive to the initial value and needs to be initialized with other methods (e.g. DLT).

⑥ Comparison with modern methods

Zhang Zhengyou calibration method: more general, supports arbitrary planar calibration plates and estimates tangential aberrations at the same time, but requires more image data.

Deep learning based calibration: no need for calibration plates, but relies on a large amount of training data, less real-time.

Applicable scenarios of RAC: still widely used in industrial vision systems (e.g., robots, drones) for calibration of lenses sensitive to radial aberrations.

(7) Practical application examples

Calibration of omnidirectional vision system

Fisheye lens calibration: using RAC method to correct the radial aberration of fisheye image, combined with external parameter calibration to realize 360° environment perception.

Multi-camera stitching system: Calibrate the internal and external parameters of multiple cameras to ensure seamless stitching (e.g., surround view system for autonomous driving).

Simplified steps

When using calibration tools (e.g. OpenCV's `calibrateCamera` function), the idea of RAC is still partially inherited, but combined with a more comprehensive aberration model (e.g. simultaneous estimation of tangential aberrations).

⑧ Improvement and Extension

Combined with master point optimization: take the master point (u_0, v_0) as the variable to be optimized to improve the calibration accuracy.

Multi-stage calibration: calibrate the internal parameter first, and then calibrate the external parameter, which is suitable for dynamic scenarios (e.g. mobile robots).

Fusion with sensors: Combine IMU (Inertial Measurement Unit) data to improve calibration robustness.

Summary

R. Tsai's RAC method is a landmark work in the field of camera calibration, especially for calibrating lenses with strong radial aberrations. Although modern methods (e.g., Zhengyou Zhang calibration method) are more general, the core ideas of RAC (e.g., step-by-step optimization, radial alignment constraints) are still important references for the design and optimization of industrial vision systems. In practical

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applications, the appropriate method needs to be selected according to the lens type (fisheye, wide-angle) and calibration requirements.

2.4 Zhang calibration method based on planar (2D) targets

In practical applications, a three-dimensional (3D) three-dimensional target production cost is high, and its processing accuracy is also subject to certain limitations, so based on this, a calibration method based on two-dimensional (2D) planar target is proposed. In this method, it is assumed that the internal parameters of the camera do not change with the movement of the camera or the target, and the parameters are calibrated by more than two images taken at different positions of the camera and the same target [14].

Among the calibration methods based on planar targets, the most classic one is the new calibration method for radial aberrations proposed by Zhang Zhengyou et al. in 1998 [22][23] - the calibration method based on planar square points, whose imaging model is shown in Figure 2-6 Zhang Zhengyou calibrated imaging model. The method uses a planar target with a large number of square grid points as calibration points, and establishes the relationship between each feature point and its corresponding image point on the image by letting the camera shoot the target from different angles to obtain two or more images. Similar to the Two-Stage calibration method, the specific steps of this method are as follows: first, the initial values of some parameters are solved by linear calculation; then, the results of the initial calculation are nonlinearly optimized according to the maximum likelihood criterion, taking radial aberration into account; and finally, the external parameters are solved with the help of a single-stress matrix and an internal reference matrix.

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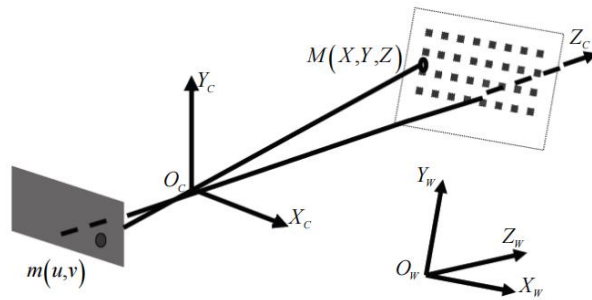


Figure 2-6 Zhang Zhengyou calibrated imaging model

(1) Mapping matrix between the target plane and its image plane.

The calibration target is shown in Figure 2-7

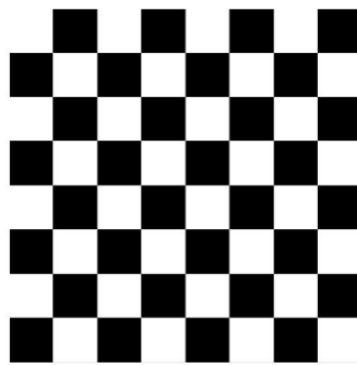


Figure 2-7 Schematic diagram of a flat chevron target

Taking a 3D feature point $M=(X,Y,Z)^T$ on a planar target, its corresponding projection point $m=(u,v)^T$ on the imaging plane, the corresponding chi-square coordinates are $M=(X,Y,Z,1)^T$ and $m=(u,v,1)^T$, which are obtained from the camera pinhole imaging model:

$$sm = A[R \ t]M \quad (2-18)$$

where $[R \ t]$ is the camera external reference; s is an arbitrary non-zero scale factor, and A is the camera internal reference matrix defined as:

$$A = \begin{bmatrix} \alpha_x & r & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2-19)$$

In this equation, (u_0, v_0) are the coordinates of the principal point; α_x, α_y denote the scale factor on the U-axis and V-axis, respectively; r is the distortion factor. Since the target is a plane, take $Z = 0$, remember the rotation matrix $R = (r_1, r_2, r_3)$, and substitute into Eq. (2-18) to obtain:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = A \begin{bmatrix} r_1 & r_2 & r_3 \\ t \end{bmatrix} \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} = A \begin{bmatrix} r_1 & r_2 & t \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \quad (2-20)$$

At this time, the feature point on the target is recorded as $M = (X, Y)^T$, $M = (X, Y, 1)^T$, and there exists a transformation matrix H between the feature point M and its projection point m. The transformation matrix H is the transformation matrix of the target:

$$sm = HM \quad (2-21)$$

In this equation, H is a 3x3 uni-responsive matrix which can be denoted as:

$$H = [h_1, h_2, h_3] = \lambda A \begin{bmatrix} r_1 & r_2 & t \end{bmatrix} \quad (2-22)$$

where r_1, r_2 are the direction vectors of the two coordinate axes of the image plane in the world coordinate system; the translation vector t is the vector from the origin of the world coordinate system to the center of the light; and λ is a constant factor. From this calculation, the matrix H is obtained.

(2) Solve the camera parameter matrix

Since the rotation matrix R has orthogonality, r_1, r_2 are satisfied:

$$\begin{cases} r_1^T r_2 = 0 \\ \|r_1\| = \|r_2\| = 1 \end{cases} \quad (2-23)$$

Obtained from equations (2-22) and (2-23):

$$\begin{cases} h_1^T A^{-T} A^{-1} h_2 = 0 \\ h_1^T A^{-T} A^{-1} h_1 = h_2^T A^{-T} A^{-1} h_2 \end{cases} \quad (2-24)$$

The above equation shows two basic constraints about the internal parameters of the camera, and there are and only two constraints for each transformation matrix.

The spatial quadratic curve can be expressed as:

$$\tilde{x}^T B \tilde{x} = 0 \quad (2-25)$$

where $\tilde{\mathbf{x}} = (X, Y, Z, 1)^T$; B is a 4x4 symmetric matrix. In expressing quadratic curves in the plane, let the above equation $\tilde{\mathbf{x}} = (X, Y, 1)^T$; B be a 3x3 symmetric matrix. In both space and plane quadratic curves, B multiplied by any non-zero scalar leaves the curve described unchanged. Therefore, $\tilde{\mathbf{x}} = (X, Y, 1)^T$ describes the projection of an absolute quadratic curve (absolute conic) onto the image plane.

Let

$$\mathbf{B} = \mathbf{A}^{-T} \mathbf{A}^{-1} = \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\alpha_x^2} & -\frac{r}{\alpha_x^2 \alpha_y} & \frac{v_0 r - u_0 \alpha_y}{\alpha_x^2 \alpha_y} \\ -\frac{r}{\alpha_x^2 \alpha_y} & \frac{r^2}{\alpha_x^2 \alpha_y^2} + \frac{1}{\alpha_y^2} & -\frac{r(v_0 r - u_0 \alpha_y)}{\alpha_x^2 \alpha_y^2} - \frac{v_0}{\alpha_y^2} \\ \frac{v_0 r - u_0 \alpha_y}{\alpha_x^2 \alpha_y} & -\frac{r(v_0 r - u_0 \alpha_y)}{\alpha_x^2 \alpha_y^2} - \frac{v_0}{\alpha_y^2} & \frac{(v_0 r - u_0 \alpha_y)^2}{\alpha_x^2 \alpha_y^2} + \frac{v_0^2}{\alpha_y^2} + 1 \end{bmatrix} \quad (2-26)$$

Since B is a symmetric matrix, the above equation can be alternatively expressed as:

$$\mathbf{b} = [B_{11}, B_{12}, B_{22}, B_{13}, B_{23}, B_{33}]^T \quad (2-27)$$

Let the i th column vector in matrix H $\mathbf{h}_i = [h_{i1}, h_{i2}, h_{i3}]^T$,

$$\mathbf{h}_i^T \mathbf{B} \mathbf{h}_j = \mathbf{v}_{ij}^T \mathbf{b} \quad (2-28)$$

In this formula:

$$\mathbf{v}_{ij} = [h_{i1}h_{j1}, h_{i1}h_{j2} + h_{i2}h_{j1}, h_{i2}h_{j2}, h_{i3}h_{j1} + h_{i1}h_{j3}, h_{i3}h_{j2} + h_{i2}h_{j3}, h_{i3}h_{j3}]^T$$

Then the above equation can be written as:

$$\begin{bmatrix} \mathbf{v}_{12}^T \\ (\mathbf{v}_{11} - \mathbf{v}_{12})^T \end{bmatrix} \mathbf{b} = 0 \quad (2-29)$$

If the system of equations obtained from the n images taken is superimposed we get:

$$\mathbf{V} \mathbf{b} = 0 \quad (2-30)$$

where V is a $2n \times 6$ matrix, if $n > 2$, then uniquely determine b in the sense of a difference of one scale factor. After obtaining the solution for b , solve A^{-1} by the Cholesky matrix factorization algorithm, and finally by equation (2-22):

$$\begin{aligned} r_1 &= \lambda A^{-1} h_1, r_2 = \lambda A^{-1} h_2 \\ r_3 &= r_1 \times r_2, t = \lambda A^{-1} h_3 \end{aligned} \quad (2-31)$$

where the scale factor $\lambda = 1/\|A^{-1}h_1\| = 1/\|A^{-1}h_2\|$. Due to the practical application, the camera lens has aberrations, so the parameters sought above are only used as initial values, after which an optimization search is carried out to calculate the exact values of the parameters.

The calibration method based on planar square points proposed by Zhengyou Zhang can be regarded as a milestone in the field of calibration. It can be adapted to the vision system built by all kinds of conventional lenses and cameras. This method is in between the self-calibration method and the traditional calibration method, and compared with the self-calibration method, it has better accuracy; compared with the traditional calibration method, it has the advantages of simplicity and convenience, and low requirements for equipment. However, there are limitations in this method, which is not suitable for non-professionals because they need to specify the physical coordinates of the feature points on the target mark and complete the matching of the points between the image and the template.

2.5 Chapter Summary

In this chapter, we discuss the imaging principles and imaging models around conventional vision systems, and analyze and compare a variety of classical calibration methods. However, fisheye lenses are designed and manufactured based on isometric

projection, isostereoscopic projection, stereoscopic projection and orthographic projection models, which makes the pinhole model difficult to be used to describe the imaging laws of fisheye lenses. Moreover, there is a significant aberration problem in such ultra-large wide-angle lenses as fisheye lenses, and the aberration parameter must be taken into account during calibration. Therefore, it is necessary to conduct an in-depth study on the parameter calibration method for the omni-directional vision system based on the fisheye lens.

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3. Fisheye lens imaging model and aberration analysis

Fisheye lenses have extremely wide fields of view and short focal lengths. Generally speaking, lenses with a field of view of more than 180° and a focal length of 16mm or less are categorized as fisheye lenses. However, from an engineering point of view, lenses with a field of view of 120° or more can also be considered fisheye lenses. This type of lens adopts the classic retro-reflective telephoto structure, which generally consists of a front lens group that assumes the imaging function and a rear lens group that is responsible for correction[11].

3.1 Structure of Fisheye Lens

Fisheye lens is a special category of ultra-wide-angle lens, the lens in front of the parabolic protruding outward, from the morphology and the fish's eyes are very similar to the visual effect produced by the fish in the water to observe the surface of the scene presented in Figure 3-1.

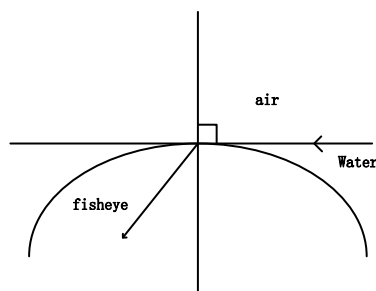


Figure 3-1 Fisheye imaging schematic

As shown in Figure 3-2, the phenomenon belongs to the principle of optics of total reflection and light path reversible.

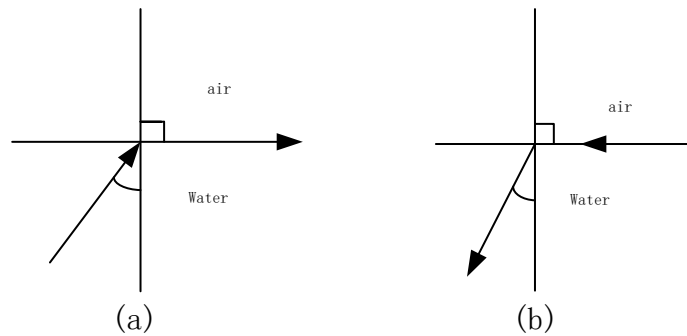


Figure 3-2 Total reflection and optical path reversibility

In Fig(a), there exists an angle of refraction i that is always greater than the angle of incidence i' , i.e., due to the refractive index $n > 1$ of water:

$$\sin i' = n \sin i \quad (3 - 1)$$

When $i' = 90^\circ$, the refracted light passes along the water surface, then the angle of incidence corresponding to it is the critical angle of total reflection, i.e.:

$$i = \arcsin \frac{1}{n} \quad (3 - 2)$$

According to the principle of reversible light path, the incident light along the water surface forms a refracted ray with refracting angle i in the water, as shown in Fig.3-2(b). It can be seen that when the fish is close to the water surface, the fish eye can gaze at the visual information in the hemispherical domain on the water surface.

If the above phenomenon in front of the convex fish eye in front of the surface and the water medium in front of its eyes as a whole as a negative lens as a fisheye front lens, as shown in Figure 3-3 (a). Because of the curvature of the surface in front of the fish eye is very large, so the front lens and the scene according to the absolute value of a large negative optical focal length, these for the optical design of the fisheye lens provides a great revelation.

Fisheye lens design idea is to use other optical materials to replace the above lens in the water medium constitutes the equivalent of the front lens optical structure, as shown

in Figure 3-3 (b), can be in other general natural conditions (such as air, etc.) to achieve the function of the fish eye in the water looking up at the water surface of the 180° scene.

Since in the structure shown in Figure 3-3(b), when the angle of the incident light from the object point has 90° , only a small amount of light in the image enters the system, and its light energy is not enough to cause the sensor to sense it. As shown in Figure 3-3(c), because the convex surface can absorb more light than the plane, so the front surface of the front lens into a convex surface, and to ensure that the original optical focal length of the lens remains unchanged, and accordingly increase the curvature of the rear surface. Fisheye lens is in the first lens after setting up a number of lenses to form a lens group to simulate the refractive process of the fisheye to ensure that the first lens beam can be accurately focused and have a good focusing quality.

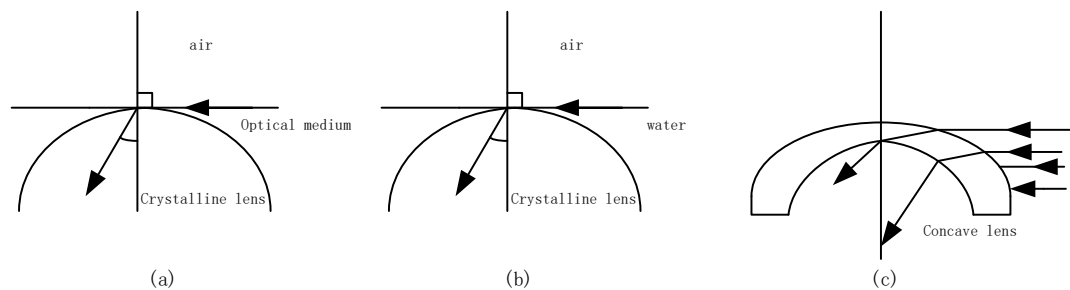


Figure 3-3 Shots obtained by fisheye elevation revelation

As shown in Figure 3-4, a common structural feature of fisheye lenses is that the front and rear groups of lenses are negatively positive in optical focal length, respectively. This feature increases the rear working distance of different types of lenses with the same focal length (greater than its focal length value). As a result, this type of lens has a wider angle of view, as well as a very short focal length.

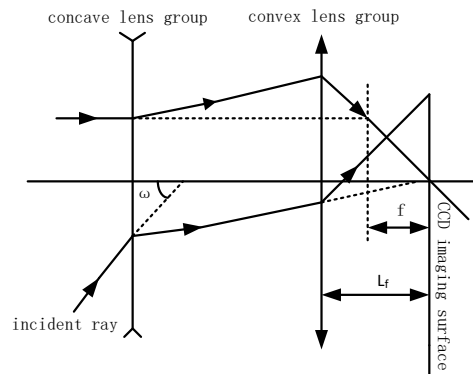


Figure 3-4 Simplified diagram of the basic structural system of the fisheye lens

3.2 Fisheye Lens Imaging Principle

Typically, the design of conventional optical imaging systems follows the principle of similarity - there is always a similarity between the image and the object - and system calibration is an attempt to refine this similarity with the following ideal imaging equation:

$$\text{Close objects: } z'_0 = \beta z$$

$$\text{Infinitely distant objects: } z'_0 f \tan \omega \quad (3 - 3)$$

Where β denotes the lateral magnification; ω denotes the object-side half-field-of-view angle; z'_0 prime denotes the ideal height; z is the object height; f denotes the optical system object-side focal length; but similar imaging cannot be realized when taking $\tan 90^\circ = \infty$.

Fisheye lens through the compression of the captured image, the introduction of deformation and realize the field of view expansion of conventional lenses, for this reason, it is necessary to choose a suitable imaging formula instead of equation (3-3) in many fisheye lens imaging model, so the selected formula has the following characteristics:

- ① The ideal imaging height described is smaller than the similar imaging.
- ② It is easy to analyze and calculate the object-image relationship.
- ③ When the range of field of view of fisheye lens is the defining domain, the ideal imaging height is a continuous and derivable function.
- ④ Familiar to the industry.
- ⑤ Simple form.

Each of the following formulas essentially meets the above conditions:

$$\begin{cases} z'_0 = 2f \tan(\omega/2) \\ z'_0 = kf\omega \\ z'_0 = 2f \sin(\omega/2) \\ z'_0 = f \sin \omega \end{cases} \quad (3-4)$$

The central goal of the above equation is to induce deformation, the corresponding curve of which is shown in Figure 3-5. In the figure, $\tan\omega$ represents the ideal imaging relation defined by Gaussian optics. We measure the amount of barrel distortion that can be introduced by presenting the difference between various imaging ideas and $\tan\omega$. These imaging ideas achieve large-angle stereo angle coverage in the object space, which is achieved with the help of varying degrees of compressive distortion of the image.

In this optical system, the path of propagation of the main light rays plays a decisive role, and it completely determines the degree of aberration produced in the system. Aberrations cause the image to deform, but this deformation has the prerequisite that the objects are no longer similar to each other without reducing the clarity of the image. From the mathematical principle, this image deformation does not destroy the one-to-

one mapping relationship between objects and images, which also ensures that the non-similarity imaging idea is feasible and correct in theory and practice.

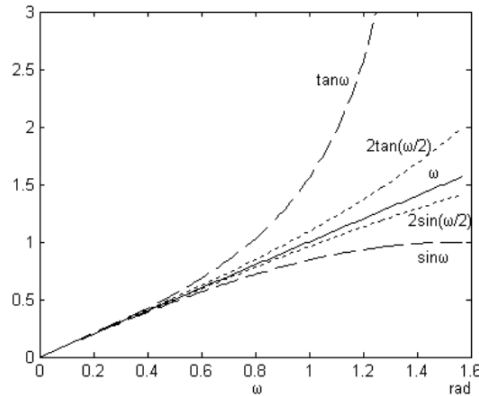


Figure 3-5 Fisheye lens imaging model curve

The imaging model for each equation is shown below:

1) “Isometric projection” imaging

$$z'_0 = f\omega \quad (3-5)$$

Where ω is the angle of incidence of the object point, differentiating the above equation gives:

$$dz'_0 = f d\omega \quad (3-6)$$

Radial magnification is obtained from the concept of magnification (Gaussian optics):

$$f\beta_r = \frac{dz'}{r d\omega} \quad (3-7)$$

Equation (3-7) is obtained by substituting (3-6):

$$\beta_r = \frac{f}{r} \quad (3-8)$$

The tangential magnification is obtained from the geometric relationship:

$$\beta_t = \frac{f}{r} \left(\frac{\omega}{\sin \omega} \right) \quad (3-9)$$

In off-center versus imaging center imaging:

$$\begin{cases} |\beta_r| < |\beta_t| & (\omega \neq 0) \\ |\beta_r| = |\beta_t| = \frac{f}{r} & (\omega = 0) \end{cases} \quad (3-10)$$

Fisheye lenses that use the design idea of equidistant projection will maintain a constant radial magnification when photographing a specific spherical object; and when there is a difference in the radius of the spherical object, the radial magnification will be different, and is in a positive proportion to the radius of curvature of the sphere.

The so-called “equidistant projection” means that the object points with equal field of view angles are presented at equal radial distances on the image plane. In general, a tiny circle on a spherical object plane will form a tiny ellipse on the image plane after equidistant projection, the long and short axes of the ellipse are along the tangential and radial directions respectively, and the size of the long axis is proportional to the field of view angle. When the field of view angle ω is equal to 0, the object near the optical axis and the image show a similar state.

The amount of aberration introduced during isometric projection imaging is capable of being adjusted artificially. In the design of the fisheye lens, enough “barrel” aberration is very critical, the specific isometric projection imaging introduced by the aberration can refer to Table 3-1.

There is a positive proportionality between the absolute value of tangential magnification β_t and the field of view angle. In the isometric projection system of thought, the image height is proportional to the field of view angle, and this property makes it easier to extract the dynamic information of changing feature points and their angular coordinates from the scene. For this reason, isometric projection has been

widely recognized as an imaging idea, and has shown great advantages in many fields of engineering and technology.

Table 3-1

field of view ω_i	0.25*0.5 π	0.45*0.5 π	0.65*0.5 π	0.85*0.5 π	1*0.5 π
$z' = f \omega_i$	0.3927f	0.7069f	1.0210f	1.3352f	1.5708f
$z'_0 = f \tan \omega_i$	0.4142f	0.8542f	1.6318f	4.1657f	∞
$\frac{z' - z'_0}{z'_0}$ relative distortion /%	-5.19	-17.24	-37.43	-67.95	-100

If there is a need to introduce a larger amount of distortion then it can be done by the following equation:

$$z' = kf\omega \quad (0 < k < 1) \quad (3 - 11)$$

where k is a constant factor, the value of which is determined by the amount of distortion required.

2) “Equal Stereo Angle Projection” Imaging

$$z' = 2f \sin\left(\frac{\omega}{2}\right) \quad (3 - 12)$$

According to the geometric principle, when the small stereo angle $d\Omega$ increases in the object space, the corresponding small surface element on the image plane will also increase, and the ratio of the two is exactly equal to the square of the focal length, i.e.: $ds' = f^2 d\Omega$.

Based on this relationship, can be derived from a simple and practical engineering measurement method: simply calculate the object side of the tiny stereo angle on the image plane corresponding to the area, you can derive the size of the stereo angle. At the same time, the tangential and radial magnification of an equirectangular projection can be derived and calculated in a similar way to that of an equirectangular projection.

$$\begin{cases} \beta_t = \frac{f}{r \cos\left(\frac{\omega}{2}\right)} \\ \beta_r = \frac{f}{r} \cos\left(\frac{\omega}{2}\right) \end{cases} \quad (3 - 13)$$

The “barrel” distortion provided by this imaging idea is slightly larger than that of an isometric projection, so this distortion is often used in civilian fisheye lenses.

3) “Stereographic projection” imaging

$$z' = 2f \tan\left(\frac{\omega}{2}\right) \quad (3 - 14)$$

In this imaging mode, tiny metasurfaces are able to maintain their shape because the radial magnification of metasurfaces at the same position has the same value as the tangential magnification. However, the metasurfaces corresponding to points at different locations on the sphere exhibit different magnifications during the imaging process:

$$\beta_r = \beta_t = \frac{f}{r \cos^2\left(\frac{\omega}{2}\right)} \quad (3 - 15)$$

So far, fisheye lenses based on this imaging idea have been used sparingly.

4) “Orthogonal projection” imaging

$$z' = f \sin \omega \quad (3 - 16)$$

The radial and tangential magnification of this imaging modality is:

$$\begin{cases} \beta_r = \frac{f}{r} \cos \omega \\ \beta_t = \frac{\beta_r}{\cos \omega} = \frac{f}{r} \end{cases} \quad (3 - 17)$$

Among the above imaging ideas, the one that introduces the largest “barrel” aberration is orthogonal projection, which makes the illumination distribution on the image surface more uniform, and therefore a few fisheye lenses use this imaging idea.

Through the above analysis, the following conclusions can be drawn:

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(1) “Similar Imaging” has the limitation of not being able to capture the ultra-wide angle information in the scene. Mere aberration only destroys the similarity between the image and the physical object, and does not lead to blurring of the image, which affects the sharpness, so it is possible to realize ultra-wide-angle imaging by sacrificing “similarity”.

(2) In order to realize ultra-wide-angle imaging in the airspace, based on the concept of “non-similar” imaging, “barrel” distortion is introduced, and the image size is compressed by using this distortion.

(3) The idea of isometric projection is widely used in common lenses. This is because according to its theorem, the image height will increase with the increase of the field of view angle, this feature facilitates information processing, meets the needs of information technology, not only real-time is good, but also high precision, so in the field of engineering and science and technology has significant advantages.

In the fisheye lens, the imaging accuracy not only depends on the model, the optical design will also have an important impact on the imaging effect. Since the fisheye lens contains multiple sets of lenses, it is difficult to achieve an ideal imaging model in the actual manufacturing and optimization design process, and problems such as optical spacing deviation, center thickness deviation, lens surface face shape error, lens eccentricity, etc., will occur. Therefore, in order to more accurately simulate the real imaging process, a reasonable aberration coefficient needs to be introduced.

3.3 Fisheye Vision System Model

3.3.1 Geometric and Mathematical Modeling of the Fisheye Visual System

The imaging mechanism of fisheye lens is significantly different from that of conventional lens. In order to accurately portray the projection position of the target point in three-dimensional space on the camera imaging plane, as well as to establish the mapping relationship between the feature points and the image pixel points, it is necessary to construct the imaging model of the fisheye vision system and explore its imaging law in depth. According to the previous discussion, the design of fisheye lens usually follows the isometric projection theorem. Therefore, in this paper, the isometric projection theorem will be used as the theoretical basis to analyze the fisheye vision system in detail. After any point in space is refracted by the fisheye lens, the imaging effect in the camera is shown in Figure 3-6.

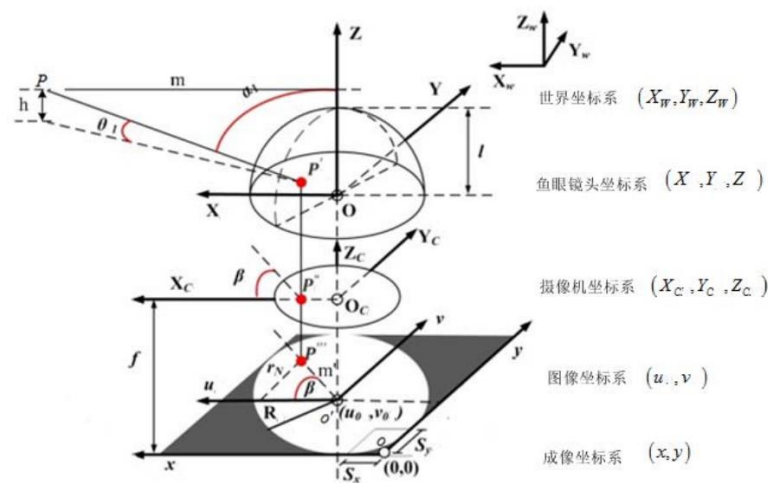


Figure 3-6 Fisheye lens imaging system model

A complete geometric imaging model of the fisheye lens omnidirectional vision system is shown in Figure 3-6. In this figure, there are a total of five coordinate systems that make up this fisheye camera imaging model. They are: the world coordinate system

(X_w, Y_w, Z_w) , the fisheye lens coordinate system (X, Y, Z) , the camera coordinate system (X_c, Y_c, Z_c) , the image coordinate system (u, v) , and the imaging coordinate system (x, y) . p is a point in the 3D scene; h denotes the perpendicular distance between the spatial point p and the projection plane; R denotes the fisheye image radius; P' is the projection point of the scene point P on the imaging plane; ω denotes the elevation angle (angle of incidence) of the point P relative to the lens center point O ; r is the radial distance between the image center point O' and the image point; θ is the azimuthal angle of the scene point P in the camera coordinate system; and S_x, S_y denote the physical dimensions of the pixels in the horizontal as well as the vertical axes, respectively. It is obtained according to the principle of isometric projection:

$$r = kf\omega \quad (3 - 18)$$

As shown in Figure 3-7, based on the equidistant projection theorem, the above complete geometric model can be reduced to a concise model describing the relationship between the angle of incidence of a field point and its radial distance in the imaging plane.

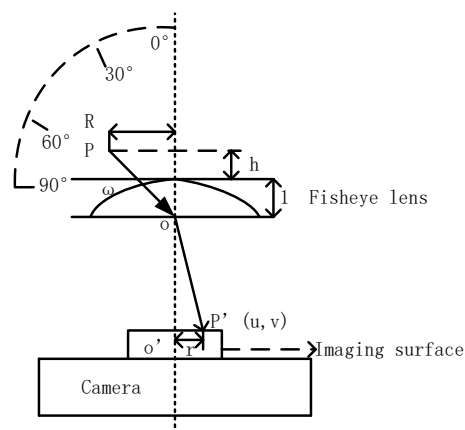


Figure 3-7 Correspondence between the radial distance of a point in the scene on the imaging plane and its elevation angle to the fisheye lens

In the figure, OO' is the optical axis of the fisheye lens; R denotes the horizontal distance between the point P and the center point O of the fisheye lens; h denotes the vertical distance between the point P and the mirror; l denotes the distance between the mirror and the imaging center point O' ; (u_0, v_0) are the coordinates of the image center point O' ; and (u, v) are the image point P' pixel coordinates in the imaging coordinate system. According to the equidistant projection theorem it is known that the same field of view angle ω corresponds to the radial distance r on the image plane, then the relation can be obtained as follows:

$$r = k\omega \quad (3-19)$$

In this equation:

$$r = \sqrt{(u - u_0)^2 + (v - v_0)^2} \quad (3-20)$$

From formula $\tan \omega = \frac{R}{h+l}$ we can get:

$$\omega = \tan^{-1} \left(\frac{R}{h+l} \right) \quad (3-21)$$

Substituting equations (3-20) and (3-21) into equation (3-19) yields:

$$K = \frac{r}{\omega} = \frac{\sqrt{(u - u_0)^2 + (v - v_0)^2}}{\tan^{-1} \left(\frac{R}{h+l} \right)} \quad (3-22)$$

3.3.2 Aberration modeling of the fisheye visual system

The root cause of lens aberrations is the camera optical imaging system in the presence of optical errors, resulting in the field of viewpoints in the camera imaging plane of the actual projection position and the ideal imaging position can not be a perfect fit. There are two main types of lens aberrations: one is radial aberrations, which

will only cause radial positional deviation; the other is tangential aberrations (also known as eccentric aberrations) and thin prism aberrations, which can cause both radial and tangential deviations [15][16].

① Radial distortion

Generally speaking, radial aberration mainly stems from the irregularity of the camera lens surface shape, resulting in the actual imaging position deviates from the ideal imaging position, which in turn causes image distortion. It is manifested in the image of the point along the radial direction to the center of the system as the center of the offset, and this offset has a positive and negative nature. When the image point to the positive direction of the offset, will lead to the image point away from the center of the image, the formation of pillow-shaped distortion, this is because with the increase in the field of view angle, the optical system of the hammer axis magnification will also increase accordingly. And when the image point is shifted in the negative direction, barrel distortion is formed. Radial aberration has a centrosymmetric property, and the farther the image point is from the center position, the more significant the degree of aberration. A schematic diagram of radial distortion is shown in Figure 3-8, where a) is a normal image without distortion, b) is a barrel distorted image, and c) is a pillow distorted image. The basic mathematical model of radial distortion is:

$$\begin{cases} \Delta x_r = x(k_1 r^2 + k_2 r^4 + k_3 r^6 + \dots) \\ \Delta y_r = y(k_1 r^2 + k_2 r^4 + k_3 r^6 + \dots) \end{cases} \quad (3-21)$$

In this equation $(\Delta x_r, \Delta y_r)$ represents the radial distortion value, k_1, k_2, k_3 are the radial distortion coefficients, and $r = \sqrt{x^2 + y^2}$ is the distance from the distortion-free ideal image coordinate point to the image center [17].

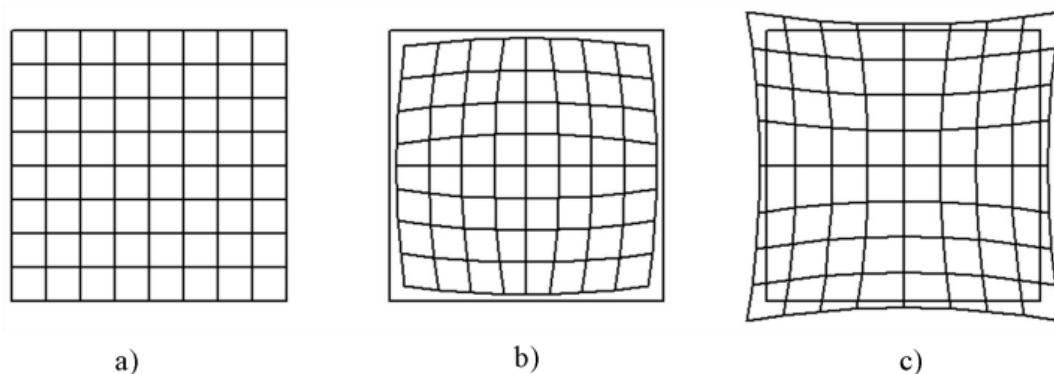


Figure 3-8 Type of radial distortion

②Centrifugal Aberration

Centrifugal aberrations are caused by a shift in the optical axis of the lens in the vision system. Tangential aberrations are mainly caused by technical constraints and insufficiently detailed processing, which prevent the optical center of the optical element from being perfectly aligned with the geometric center. Generally speaking, centrifugal aberrations tend to have less impact than radial aberrations, but in order to guarantee the accuracy of the description of the imaging process, they are still taken into account in the establishment of the imaging model, which is as follows:

$$\begin{cases} \Delta x_t = p_1(2x^2 + r^2) + 2p_2xy + \dots \\ \Delta y_t = 2p_1xy + p_2(2y^2 + r^2) + \dots \end{cases} \quad (3-22)$$

In this equation, $(\Delta x_t, \Delta y_t)$ represents the centrifugal aberration value, and p_1, p_2 are the centrifugal aberration coefficients.

Centrifugal aberration consists of radial and axial deviation components. Among them, the radial direction of the deviation from the camera and the sensor there is a radial position offset, resulting in the camera optical axis and the imaging center axis does not coincide; axial direction of the deviation is due to the camera's internal optics in the installation and commissioning process of the phenomenon of different axes,

which triggered centrifugal aberrations, ultimately leading to the camera to shoot the target when the image image of the position of the point of deviation.

③Thin Prism Aberration

Thin prismatic aberration usually stems from human intervention factors in the manufacturing process of fisheye lens, covering lens design, optical component processing and lens mounting and other process steps. In the imaging process, the imaging deviation caused by this aberration is much smaller than that of radial and tangential aberrations. Since the inclusion of its influence will lead to model parameter redundancy, which increases the complexity of the solution and even triggers the instability of the solution, the thin prism aberration can usually be ignored in the study of the camera imaging process. The mathematical model can be expressed as:

$$\begin{cases} \Delta x_p = s_1(x^2 + y^2) + \dots \\ \Delta y_p = s_2(x^2 + y^2) + \dots \end{cases} \quad (3-23)$$

In this equation, $(\Delta x_p, \Delta y_p)$ represents the centrifugal distortion value and s_1, s_2 are the centrifugal distortion coefficients.

In summary, the mathematical expressions of the above three aberration models are in the form of infinite series, which means that theoretically each aberration corresponds to an infinite number of aberration coefficients. However, in practical modeling, it is obviously not feasible to solve the computational process of the infinite terms directly, so this kind of theoretical expressions cannot be directly applied to engineering calculations. By analyzing the level structure, it can be seen that the higher terms are higher order infinitesimals compared to the primary terms, and their influence on the overall expression decreases sharply as the number of times increases. In addition, if too

many unknown parameters are introduced into the modeling of the visual system, it will not only increase the complexity of the solution, but also may lead to unstable numerical solutions or even reduce the accuracy of the results. Therefore, in practical applications, the aberration model is usually simplified: the higher-order infinitesimal terms are ignored, and only the first few terms of the infinite series are used as the approximate expression.

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4. Calibration methodology design for OmniVision

4.1 General framework of the calibration methodology

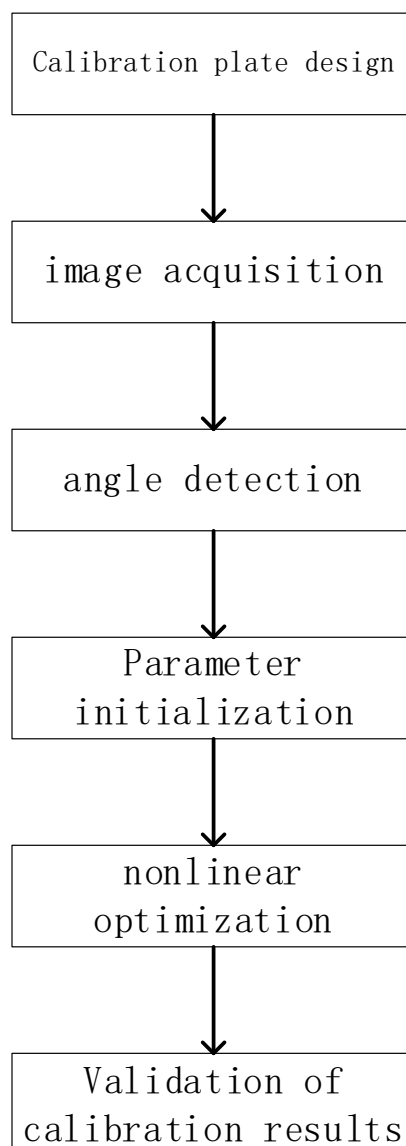


Figure 4-1 Flow chart of calibration process

1. The role of the checkerboard calibration board

Geometric constraints: The corner points of the checkerboard grid provide known 3D-2D correspondence points, and the set of overdetermined equations is established by multiple images with different angles.

Parameter design: CHECKERBOARD=(9,6) indicates that the number of inner corner points is 9 columns and 6 rows (the actual grid number is 10x7), which needs to be strictly matched with the actual calibration board.

Physical size: size_per_grid maps pixel coordinates to real world coordinates (units need to be consistent with subsequent application scenarios).

2. 3D world coordinate generation

```
objp[0,::2] = np.mgrid[0:CHECKERBOARD[0], 0:CHECKERBOARD[1]].T.reshape(-1, 2) * size_per_grid
```

Figure 4-2 code segment

np.mgrid: generate grid coordinates, e.g. x=0,1,...,8, y=0,1,...,5.

reshape(-1,2): convert coordinates to an array of (54,2), corresponding to 54 corner points.

* size_per_grid: multiply the normalized coordinates by the actual size (e.g. 116mm) to give physical meaning.

Corner point detection and optimization

cv2.findChessboardCorners:

Use adaptive thresholding (CALIB_CB_ADAPTIVE_THRESH) to cope with uneven lighting.

Fast check (CALIB_CB_FAST_CHECK) to improve detection speed.

Normalized image (CALIB_CB_NORMALIZE_IMAGE) enhances robustness.

cv2.cornerSubPix:

Optimizes corner coordinates based on gradient iteration with a search window of (3,3) and iteration termination conditions defined by subpix_criteria.

Improves subpixel level accuracy (typically up to 0.1 pixel level)

3. Fisheye calibration function parameters

```
cv2.fisheye.calibrate(
    objpoints,
    imgpoints,
    gray.shape[::-1],
    K,
    D,
    rvecs,
    tvecs,
    calibration_flags,
    (cv2.TERM_CRITERIA_EPS+cv2.TERM_CRITERIA_MAX_ITER, 30, 1e-6)
)
```

Figure 4-3 code segment

Key parameters:

objpoints: 3D points under the world coordinate system (list, each element corresponds to a corner point of a map).

imgpoints: 2D points under the image coordinate system (one-to-one correspondence with objpoints).

K: output internal reference matrix in the format:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (4 - 1)$$

D: fisheye distortion coefficient, 4 parameters (k1, k2, k3, k4).

calibration_flags: combination of flag bits that control the optimization process.

Flag bit details:

CALIB_RECOMPUTE_EXTRINSIC: force recalculation of the outer parameters (improve accuracy).

CALIB_CHECK_COND: check the matrix condition number to avoid pathological solutions.

CALIB_FIX_SKEW: simplifies the model by assuming that the image is not skewed (i.e., $s=0$).

4. Fisheye distortion model

Fisheye cameras use the Equidistant Projection model with the aberration equation:

$$r = \theta(1 + k_1\theta^2 + k_2\theta^4 + k_3\theta^6 + k_4\theta^8) \quad (4 - 2)$$

Where:

θ is the angle between the incident light and the optical axis.

r is the distance of the image point from the principal point (in pixel units)

4.2 Calibration Process Summary and Parameter Interpretation

1. Summary of the calibration process

① Data Acquisition: Take images of the checkerboard grid from different angles (15-20 images are recommended to cover the whole field of view).

② Corner point extraction: Detect the corner points in each image and record their corresponding 3D world coordinates.

③ Parameter initialization: the internal reference matrix K is initialized to zero, and the distortion coefficient D is initialized to zero.

④ Nonlinear optimization: minimize the reprojection error by the Levenberg-Marquardt algorithm, solve for K and D .

⑤ Validation of results: Check the RMS error (typically <0.5 pixels is a good result) and validate the calibration by visualizing the de-distorted images.

2. Interpretation of the output parameters

DIM: Image size (width x height) for subsequent de-distortion.

K: internal reference matrix containing focal length (f_x , f_y) and principal points (c_x , c_y).

D: distortion factor, e.g. $D=[0.1, -0.02, 0.003, 0.0001]$.

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5. Experiments and analysis of results

1. First select the fisheye camera image (generated by Unity) to be calibrated as follows Figure 5-1:

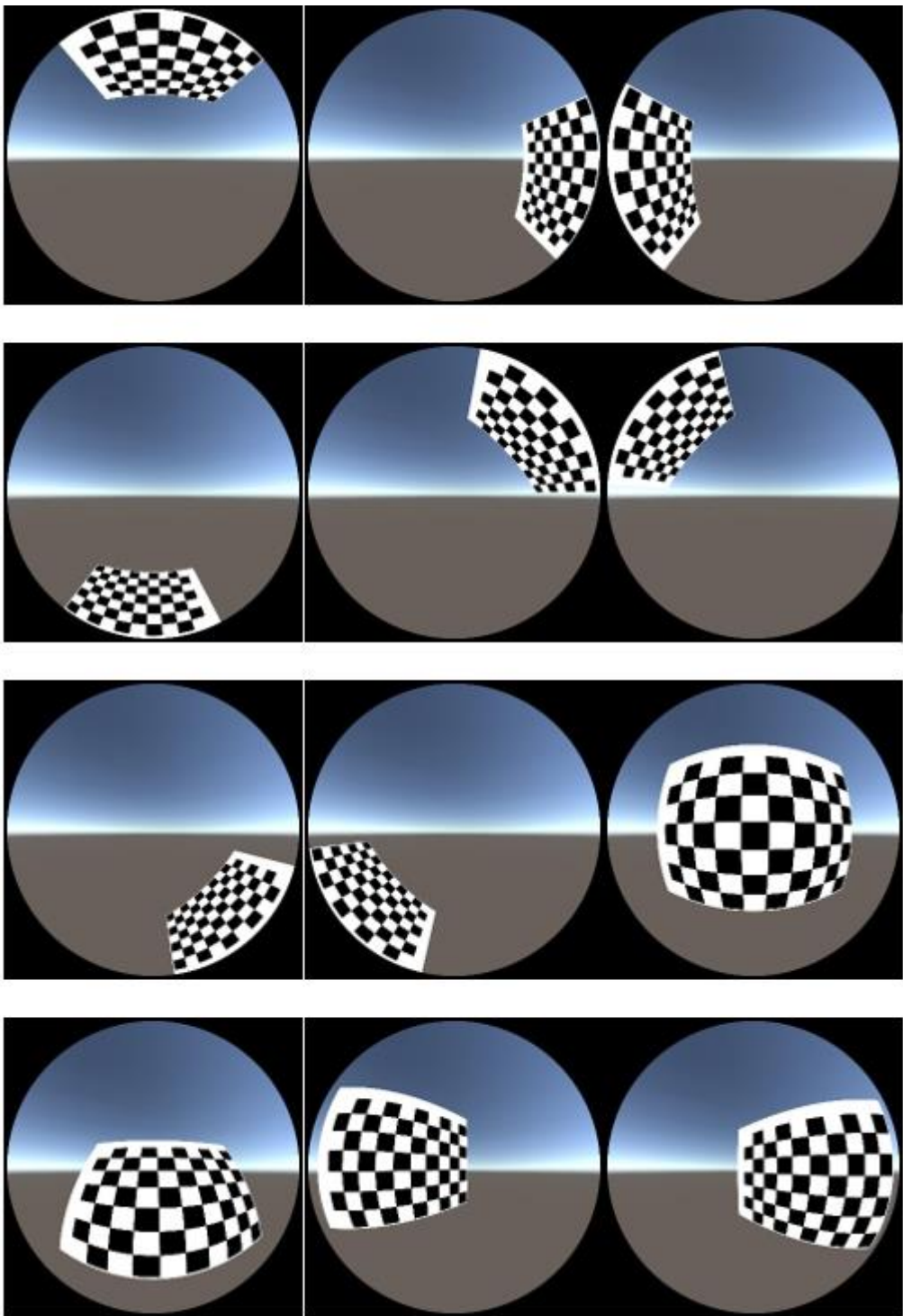


Figure 5-1 Calibration images

2. Calibrate the Python code after filling in the correct path for saving the image

Below is part of the code in Figure 5-2 :

```
import cv2
assert cv2.__version__[0] == '4', 'The fisheye module requires opencv version >= 3.0.0'
import numpy as np
import glob

CHECKERBOARD = (9,6) # 棋盘格角点数
size_per_grid = 116 # 棋盘格子尺寸
subpix_criteria = (cv2.TERM_CRITERIA_EPS+cv2.TERM_CRITERIA_MAX_ITER, 30, 0.1)
calibration_flags = cv2.fisheye.CALIB_RECOMPUTE_EXTRINSIC+cv2.fisheye.CALIB_CHECK_COND+cv2.fisheye.CALIB_FIX_SKEW
objp = np.zeros( shape: (1, CHECKERBOARD[0]*CHECKERBOARD[1], 3), np.float32)
objp[0,:, :2] = np.mgrid[0:CHECKERBOARD[0], 0:CHECKERBOARD[1]].T.reshape(-1, 2) * size_per_grid
```

Figure 5-2 part of code

The following Figure 5-3 shows the calibration results:

```
D:\doubao\.venv\Scripts\python.exe D:\doubao\fisheye_calibration.py
Found 12 valid images for calibration
DIM=(1920, 1920)
K=np.array([[610.6938947728503, 0.0, 959.7314124388965], [0.0, 610.5677694496611, 959.7389316997021], [0.0, 0.0, 1.0]])
D=np.array([[0.001698329277591275], [-0.002884210649590748], [0.001992752522615301], [-0.0004896599361001555]])
校准成功! RMS误差: 0.28237418022915173

进程已结束, 退出代码为 0
```

Figure 5-3 Calibration Results

Textual form:

Found 12 valid images for calibration

DIM=(1920, 1920)

$K = \text{np.array}([[610.6938947728503, 0.0, 959.7314124388965],$
 $[0.0, 610.5677694496611, 959.7389316997021],$
 $[0.0, 0.0, 1.0]])$

$D = \text{np.array}([[0.001698329277591275],$
 $[0.002884210649590748],$
 $[0.001992752522615301],$
 $[-0.0004896599361001555]])$

Calibration successful! RMS Error: 0.28237418022915173

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3. Calculate the distance from the origin of the two checkerboard grids to the camera



Figure 5-4 The left chessboard



Figure 5-5 The front chessboard

After calibrating the camera ,we will use the parameters to calculate the next step.Here are some part of the code in Figure 5-6 .

```
> import ...  
  
# 相机内参矩阵 K 和畸变系数 D  
K = np.array( object: [[610.6938947728503, 0.0, 959.7314124388965], [0.0, 610.5677694496611, 959.7389316997021], [0.0, 0.0, 1.0]]  
  
D = np.array( object: [0.001698329277591275,-0.002884210649590748, 0.001992752522615301,-0.0004896599361001555], dtype=np.float32  
  
# 棋盘格尺寸  
CHECKERBOARD = (9,6)  
per_grid_size = 116
```

Figure 5-6 Code part

Here are the results of the experiment in Figure 5-7

```
test_image1.jpg The distance of the origin of the checkerboard grid from the camera is: 673.6456 mm
test_image2.jpg The distance of the origin of the checkerboard grid from the camera is: 1224.8236 mm
```

Figure 5-7 Results of the distance

Now we compare the results obtained with those obtained in matlab in Figure 5-8.

```
Starting robust non-linear refinement

ss =

    1.0e+02 *

   -2.549217000280450
                   0
    0.000013829763987
   -0.000000006144804
    0.000000000028159

Root mean square[pixel]:  0.162465
```

Figure 5-8 Parameters in Matlab

From the figure , we can get the reprojection error of 0.16, which is smaller than the reprojection error of 0.28 obtained from the calibration performed by Python, indicating that the parameters obtained from the calibration of the Matlab toolbox are more accurate, the model fit is better, and the robustness and stability are stronger.

```
Processing image TestImages/image.jpg...Done

Y1 =

   -1.222896019765491e+03

Processing image TestImages/image1.jpg...Done

Y2 =

   -6.724031405260489e+02|
```

Figure 5-9 Distances of two chessboard

The above figure Figure 5-9, is the result calculated in Matlab, Y1 represents the distance from the center point of the upper checkerboard grid to the Y-axis direction of the camera, Y2 represents the distance from the center point of the left checkerboard grid to the X-axis direction of the camera.

The following is a comparison of the measured value with the true value in Matlab

	Left Chess, mm	Up Chess, mm
Real values	675.2	1223.0
Experiment values	672.4	1222.9
Error (Real – Experiment)	2.8	0.1

Figure 5-10 Comparison in Matlab

The following is a comparison of the measured value with the true value in Python

	Left Chess,mm	Up Chess,mm
Real Values	675.2	1223.0
Experiment Values	673.6	1224.8
Error(Real-Experiment)	1.6	-1.8

Figure 5-11 Comparison in Python

Experimental Design:

Calibration parameters were validated by calculating the distance from the checkerboard origin to the camera. Both MATLAB and Python produced reasonable results, but with differing error distributions:

MATLAB: Uniform overall error but larger deviation (2.8 mm) for the left checkerboard, possibly due to insufficient edge-field distortion modeling.

Python: Smaller error for the left checkerboard (1.6 mm) but slightly larger error for the upper checkerboard (-1.8 mm), reflecting varying adaptability to different viewing angles.

4. Calculate the distance from the camera to the cube in the specified direction.

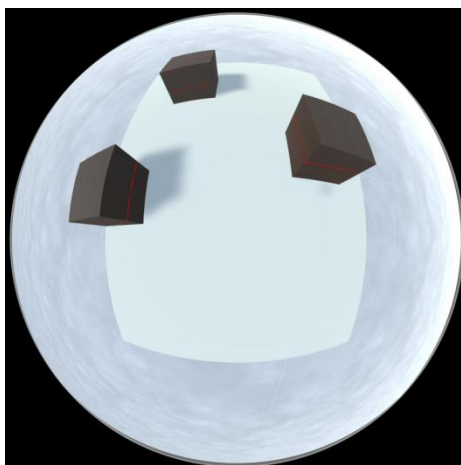


Figure 5-12 The cubes in Unity

These are some part of the main code below.

```
image_path = "0./basic/laser/image.jpg"
x_angle = 1.5
y_angle = -2.5
las_dist = 950
CVsyst_x = 0
CVsyst_y = 0

# === 加载图像和模型 ===
img_bin = las_segm(image_path)
print_laser_bounds(img_bin)

ocam_model = get_virtual_ocam_model()

# === 计算整体映射点 (可视化) ===
x1, y1 = mapping(img_bin, x_angle, y_angle, las_dist, ocam_model)

# === 模拟深度估计 ===
C_left = mock_cube_dist(img_bin, row_range: [890, 940], col_range: [350, 600], name="Left Cube")
C_Up = mock_cube_dist(img_bin, row_range: [250, 300], col_range: [700, 950], name="Up Cube")
C_Right = mock_cube_dist(img_bin, row_range: [500, 570], col_range: [1200, 1500], name="Right Cube")

# === 打印估计结果 ===
print(f"Left Cube Distance Estimate: {C_left:.2f} mm")
print(f"Up Cube Distance Estimate: {C_Up:.2f} mm")
print(f"Right Cube Distance Estimate: {C_Right:.2f} mm")
```

Figure 5-13 Code Part

Now we can compare our input image with the output one. As you can see, output image contains only distance information of the laser strip intersection with the obstacles:

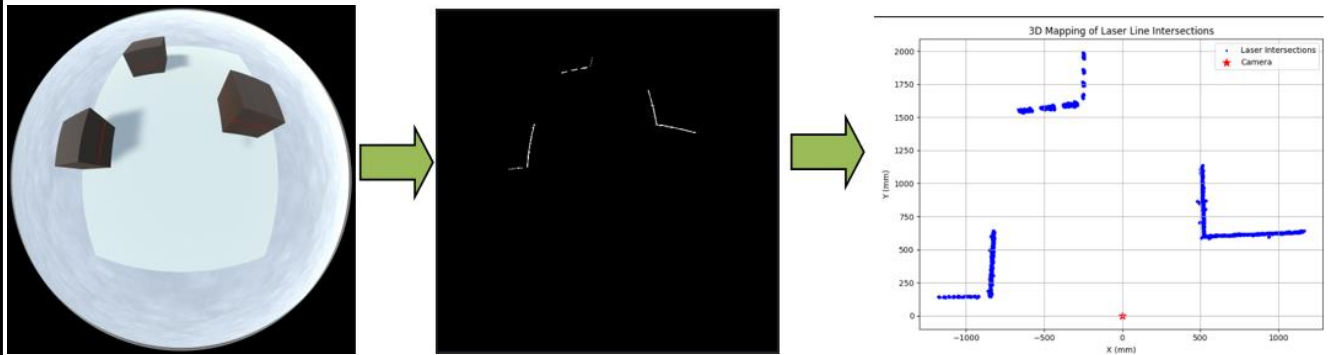


Figure 5-14 The processing

The following is a comparison of the measured value with the true value in Matlab

	Left Cube, mm	Up Cube, mm	Right Cube, mm
Real values	-922.00	1799.00	521.00
Experiment values	-921.22	1799.40	519.00
Error (Real – Experiment)	0.78	0.40	2.00

Figure 5-15 Comparision in Matlab

The following is a comparison of the measured value with the true value in Python

```

===== Error Analysis Table =====
Left Cube | Real: -922.00 mm | Predicted: -922.02 mm | Error: 0.02 mm
Up Cube   | Real: 1799.00 mm | Predicted: 1799.02 mm | Error: 0.02 mm
Right Cube | Real: 521.00 mm | Predicted: 520.98 mm | Error: 0.02 mm

```

Figure 5-16 Comparision in Python

Cube Distance Estimation:

Python achieved sub-millimeter errors (<0.1 mm) for left, upper, and right cubes in simulated depth estimation, indicating:

Calibration parameters are effective for 3D reconstruction tasks.

Python's lightweight implementation (e.g., custom laser line segmentation) may offer real-time advantages on edge devices.

6. CONCLUSIONS AND PROSPECTS

Conclusions:

1. Fisheye lenses' strong nonlinear aberrations (especially radial distortion and wide-field coupling) render traditional pinhole-based calibration methods insufficient, necessitating specialized models (e.g., equidistant projection) and nonlinear optimization algorithms.
2. The checkerboard-based calibration method, combined with nonlinear optimization (e.g., Levenberg-Marquardt algorithm), effectively solves fisheye lens parameters. However, tool-specific constraints (e.g., MATLAB vs. Python) influence accuracy: MATLAB achieves lower reprojection error, while Python shows comparable distance-measurement accuracy after parameter tuning.
3. Experimental validation confirms the methodology's effectiveness, with Python-based results closely matching real-world values despite higher inherent errors, highlighting open-source tools' potential with fine-tuning.

Prospects:

1. Real-Time Self-Calibration: Develop dynamic calibration algorithms without physical targets to adapt to parameter drift in vibrating scenarios (e.g., autonomous vehicles, drones).
2. Multi-Sensor Fusion: Investigate joint calibration of fisheye lenses with LiDAR, IMUs, and other sensors to improve spatio-temporal alignment of multimodal data.
3. Lightweight Deep Learning Models: Design edge-deployable, end-to-end calibration networks with interpretable AI to address the physical interpretability gap in traditional deep learning approaches.

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